1. Relationship between distance and steps: the relationship between distance (d) and steps (m) follows: $d^2 \approx m$

2. Evidence

```
RandomWalkTest (cc 277 ms    ✓ 9 tests passed  9 tests total, 277 ms
  ✓ testDistanceAfterN 75 ms  /Users/liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...
  ✓ testRandomWalk2 17 ms
  ✓ testMove0 2 ms            Process finished with exit code 0
  ✓ testMove1 3 ms
  ✓ testMove2 3 ms
  ✓ testMove3 3 ms
  ✓ testRandomWalk 172 ms
```

```
/Users/liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...
50 steps: 6.2524555779984565 over 10 experiments
```

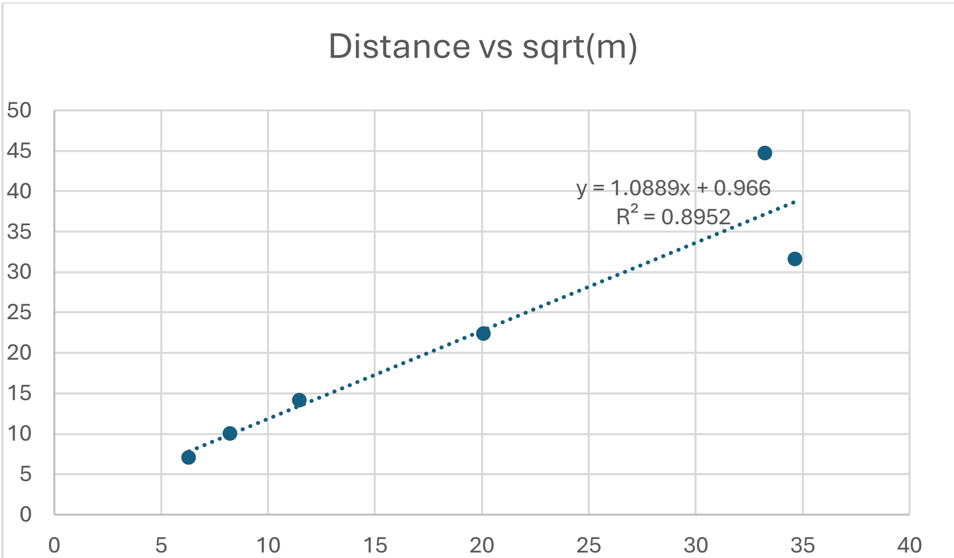
```
/Users/liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...
100 steps: 10.209699887111835 over 10 experiments
```

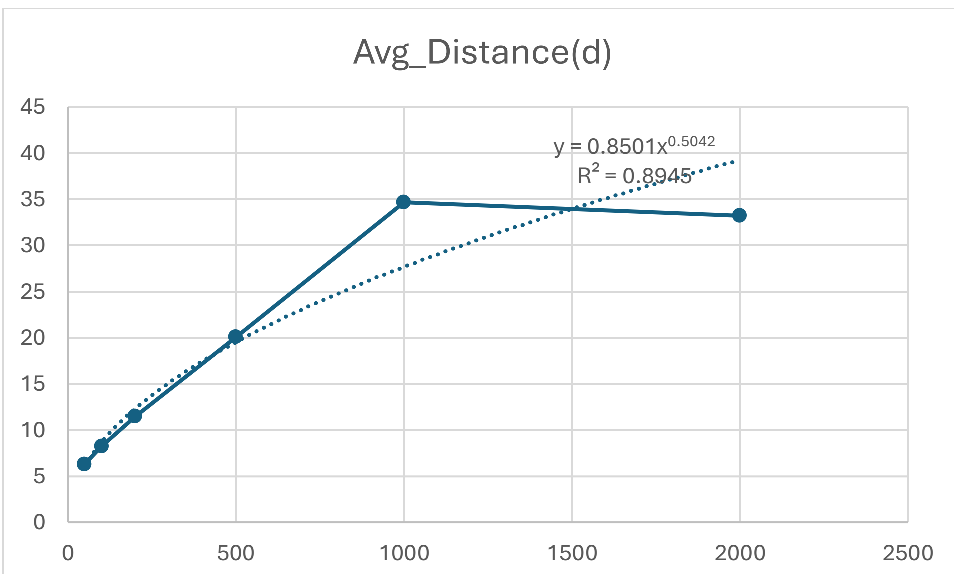
```
200 steps: 10.598426647528324 over 10 experiments
```

```
500 steps: 24.045325140081747 over 10 experiments
```

```
/Users/liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...
1000 steps: 23.735579864000158 over 10 experiments
```

```
/Users/liang.zim/Library/Java/JavaVirtualMachines/ms-21.0.9/Contents/Home/bin/java ...
2000 steps: 33.98827702866831 over 10 experiments
```





3.The expected distance from the origin increases proportionally to the square root of the number of steps($d^2 \approx m$). this means that to double the distance, the drunkard must take about four times the number of steps.